Annexure-C

ADVISORY ON THE USE OF DEEPSEEK FOR DEFENCE PERSONNEL

1. **Aim** The advisory is to sensitise all defence personnel regarding the implications of DeepSeek AI, a large language model (LLM), in terms of cybersecurity and national security.

2. **Backqround.** DeepSeek is a Large language model trained on massive datasets to understand and generate reply akin to human response, developed by Chinese AI company specialising in natural language processing and artificial intelligence. It aims to provide capabilities similar to models like OpenAI's 'ChatGPT', Google" Gemini

3. DeepSeek is an open-source software created as a part of China's broader efforts to develop independent Al models. It is designed for various natural language processing tasks like text generation, summarisation, and coding assistance. DeepSeek models have been trained on a vast data set including multi-lingual data with focus on Chinese and English.

4. **Information Security Concerns.**

(a) **Loqin Concerns**. DeepSeek allows users to login using a new account linked to an email address, Google (Gmail) credentials or a +86-phone number. The Gmail login option, while most convenient, comes with additional risks; as follows: -

(i) **Potential Exposure of Goole Account Data.** Exposure of user profile details linked contacts or login metadata.

(ii) **Security Risks.** In case of a data breach, attackers could use stolen access tokens to compromise Gmail-linked accounts.

(b) **Data Collection and Privacy Issues**. DeepSeek's privacy policy indicates that all user inputs including text, audio, uploaded files, and chat history are collected and stored on servers located in the People's Republic of China. This centralised data storage raises multiple concerns: -

(i) **Broad Data Collection.** The platform collects extensive user data that can include personal identifiers, keystroke patterns, device information, and potentially sensitive operational data.

(ii) **Lack of Data Minimisation** Similar to other Al applications, DeepSeek processes user inputs for model improvement

(c) **Potential Vulnerabilities**. There are credible concerns that DeepSeek may incorporate mechanisms, intentional or otherwise that allow for covert data access. Key factors include: -

(i) **Mandated Data Sharing under Chinese Law**. Chinese national laws (e.g., the National Intelligence Law of 2017) can compel companies to share user data with state security agencies upon request. This legal framework increases the risk that DeepSeek might be exploited as an entry point for unauthorised data access.

(ii) **Biased and Self-Censorship Practices**. DeepSeek, like other Chinese AI models, implements self-censorship on topics considered sensitive within China. This raises concerns regarding the objectivity and reliability of the information provided by the platform. The platform's integrated censorship mechanisms can interfere with data integrity and transparency, and they raise additional concerns regarding spreading of Chinese propaganda.

(iii) **International (Reports of Vulnerabilities.** Recent actions in the US and Europe, including bans on DeepSeek on government-issued devices highlight apprehensions that the platform's design may allow for unauthorised access to user data by adversarial state actors.

(d) **Comparison of DeepSeek and OpenAI wrt to Security Features.** DeepSeek's open weight models including - DeepSeek R1, DeepSeek-Coder and DeepSeek Chat are some of the most powerful AI models to date. These are open-source models which allows users to download, modify and deploy on their own systems. Unlike the traditional models like OpenAI's GPT-4o, Google's Gemini 1.5 are task specific. A comparative assessment of DeepSeek vs OpenAI with respect to compliance with security parameters is as mentioned below: -

Ser	Feature	DeepSeek	ChatGPT
(a)	Data Storage Location	China	Unspecified, but adheres to global data protection
(b)	Government Access Risks	Subject to China's strict data laws	No explicit government accesses requirements.
(c)	Data Scope Collection	Extensive user input, device data, behavioural tracking	User input technical information but with more transparency
(d)	User Control over Data	Limited; no clear opt-out mechanism.	Offers deletion, opt outs for model training.
(e)	Compliance with Privacy laws	Unclear; no explicit GDPR/CCPA mention	Explicit compliance with GDPR and CCPA

5. **Recommendations.** DeepSeek AI's data collection practices which include transmitting user data to China, raise serious privacy concerns. The use of this application by Defence personnel could lead to the compromise of personally identifiable information (PII) and other sensitive data. There, based on the above findings the following is recommended: -

(a) **Prohibition on Use:** Use of DeepSeek platform on any official, operational or sensitive devices by IN personnel is strictly prohibited.

(b) **Do Not Input Sensitive Data:** Under no circumstances should any classified, strategic, or sensitive operational data be ingested on DeepSeek platform.

(c) **Awareness and Training:** All personnel should be briefed on the potential security risks associated with using platforms subject to foreign legal regimes that mandate extensive data sharing.

(d) **Use of Alternetive AI Platforms.** Consider using AI platforms which comply with data protection regulations.

6. **Conclusion.** The combination of extensive data collection practices, potential backdoor vulnerabilities, and the legal obligations imposed on Chinese Companies to share data with the state make DeepSeek a significant security risk. As a matter of operational security, the use of DeepSeek is prohibited. Personnel must ensure that no sensitive or classified information is exposed Via this platform.

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